Artificial Intelligence

A.Y. 2025/2026

Prof. Fabio Patrizi

General Information

General Information

Instructor: Prof. Fabio Patrizi (www.diag.uniroma1.it/patrizi)

Schedule (Room 41 SPV):

- Tuesday 16:00-18:00
- Friday 12:00-15:00

Course Website: https://sites.google.com/diag.uniroma1.it/ai-25-26/home

Classroom: to get the code, fill the questionnaire

Student hour: see Instructor's website (*Teaching* section)

General Information

Starting this year:

AI (this course): 6CFU

ML (other course, with Prof. Fusco): 6CFUs

Exam

Written

Theoretical and practical questions (usually 4)

Max: 32 (yes, there are two bonus points)

Cum Laude: ≥ 31

Preliminary Program

- Introduction
- Search Algorithms (review): Uninformed and Informed Search Strategies. Heuristic Search.
- Propositional Logic (review): syntax and semantics; evaluation, satisfiability, validity, logical implication; propositional tableaux; DPLL.
- First-Order Logic (review): syntax, semantics; evaluation, satisfiability, validity, logical implication; FO tableaux.
- Reasoning about Action: modeling discrete dynamic domains, action preconditions, effects, the frame problem, observability, determinism, stochasticity.
- Classical Planning: deterministic domains, PDDL (STRIPS, ADL), Heuristics for Planning
- Contingent Planning (FOND): nondeterministic domains, PDDL with oneof operator, nondeterministic planning by adversarial search, search in AND-OR Graphs
- Knowledge and Reasoning under Uncertainty: MDPs, Value Iteration, Policy Iteration
- Situation Calculus: precondition and successor-state axioms; situation tree; regression; executability and projection.

Homework #0

Fill questionnaire ASAP! (and get the Classroom code)